

COURSE TITLE : FABRIC FORMATION II
COURSE CODE : 4061
COURSE CATEGORY : B
PERIODS/WEEK : 4
PERIODS/SEMESTER : 56
CREDITS : 4

TIME SCHEDULE

Module	Topics	Periods
1	Secondary & Auxiliary motions	14
2	Dobby Shedding	14
3	Jacquard Shedding .	14
4	Computer aided Textile Designs,Dobby & Jacquard designs	14
TOTAL		56

COURSE OUTCOMES

Module	GO	Student will be able to
1	1	Understand secondary motions of power loom
	2	Know auxiliary motions of power loom
2	1	Understand doobby shedding mechanisms & various types of bobbies.
	2	Know about Keighley doobby ,climax doobby and cross border doobby.
	3	Understand about Ruti Cam doobby, Knowles positive open shed doobby & electronics doobby
3	1	Know about jacquard shedding & the terms related to jacquard.
	2	Understand single lift, double lift & crossborder jacquards.
	3	Know about jacquard design development, card cutting, card lacing & special jacquard.
4	1	Understand application of CAD in textile designing
	2	Know the developing of doobby & jacquard designs.

SPECIFIC OUTCOME

MODULE I SECONDARY & AUXILIARY MOTIONS

1.1.0 Understand secondary motions of power loom

- 1.1.1 State the objectives of secondary motions.
- 1.1.2 Classifications of take up motions.
- 1.1.3 Explain the working of 7- wheel take up motion with sketch.
- 1.1.4 Describe the working of a continuous worm and worm wheel type take up motion with sketch.
- 1.1.5 Estimate dividend, required standard wheel & change wheel.
- 1.1.6 Classifications of let off motions.
- 1.1.7 Describe the working of a negative let off motion with sketch.
- 1.1.8 Explain the working of Ruti & Roper let off motions with sketch.

1.2.0 Know auxiliary motions of power loom

- 1.2.1 State the objectives of auxiliary motions.
- 1.2.2 Objective of weft stop motions
- 1.2.3 Distinguish side weft fork and centre weft fork motions.
- 1.2.4 Illustrate the working of side weft fork motion with sketch.
- 1.2.5 Explain the features of centre fork motion.
- 1.2.6 List the types of warp stop motions.
- 1.2.7 Explain the working of a mechanical warp stop motion used in 'Sakamoto' looms
- 1.2.8 Explain the features of electrical warp stop motion
- 1.2.9 List warp protecting motions
- 1.2.10 Explain the working of loose reed & fast reed motions with neat sketches.
- 1.2.11 Illustrate Loom brake, check strap & temples.
- 1.2.12 State the timing and setting of auxiliary motions.

MODULE II DOBBY SHEDDING

2.1.0 Understand dobby shedding mechanisms & various types of bobbies.

- 2.1.1 State the scope of Dobby shedding mechanism
- 2.1.2 Classify dobby shedding
- 2.1.3 Discuss dobby designs
- 2.1.4 Illustrate the method of pegging of lags for L.H dobby & RH dobby.

2.2.0 Know about Keighley dobby ,climax dobby and crossborder dobby

- 2.2.1 Describe Keighley dobby
- 2.2.2 Explain the working of Climax dobby.
- 2.2.3 List the parts of Climax dobby.
- 2.2.4 Explain the features & working of Cross border dobby.

2.3.0 Understand about Ruti Cam dobby, Knowles positive open shed dobby & electronics dobby

- 2.3.1 Explain the working of Ruti Cam dobby

- 2.3.2 Describe the working of Knowles positive open shed dobby.
- 2.3.3 Explain the features of Electronics dobby.
- 2.3.4 State the functions of heald leveling device and steadying mechanism in dobby

MODULE III JACQUARD SHEDDING

3.1.0 Know about jacquard shedding & the terms related to jacquard.

- 3.1.1 State the basic principles of Jacquard shedding.
- 3.1.2 List the various classifications of Jacquards
- 3.1.3 Illustrate various types of harness ties
- 3.1.4 Define terms - count of comber board, First hook and First needle of the jacquard.
- 3.1.5 Illustrate harness mounting London and Norwich systems.

3.2.0 Understand single lift, double lift & crossborder jacquards.

- 3.2.1 Illustrate the working of Single Lift Jacquard with neat sketch
- 3.2.2 Describe the working of Double Lift Single Cylinder Jacquard
- 3.2.3 Explain the working of Double Lift Double Cylinder Jacquard
- 3.2.4 Illustrate the working of Cross Border Jacquard

3.3.0 Know about jacquard design development, card cutting, card lacing & special jacquard.

- 3.3.1 Describe the construction and development of Jacquard designs.
- 3.3.2 Explain the working of Piano card cutting machine.
- 3.3.3 State the method of card punching and card lacing.
- 3.3.4 Illustrate the Special Jacquards – Pressure Harness and Bannister Harness.

MODULE IV COMPUTER AIDED TEXTILE DESIGNS, DOBBY & JACQUARD DESIGNS

4.1.0 Understand application of CAD in textile designing

- 4.1.1 State the use of computer aided textile designing & lists the software's
- 4.1.2 List the common features of Textile CAD
- 4.1.3 Describe the creation of weave, draft and peg plan by using CAD
- 4.1.4 Illustrate the Creation of weave library
- 4.1.5 State the steps to produce stripe, cross over and check designs
- 4.1.6 State the steps to produce a printed design and its colour Separations.
- 4.1.7 Explain the fabric simulation

4.2.0 Know the developing of dobby & jacquard designs.

- 4.2.1 Identify the steps to produce a dobby design by using CATD.
- 4.2.2 State steps & commands to produce a Jacquard design by using CATD
- 4.2.3 Explain fabric simulation with different Colour combinations.
- 4.2.4 Define the shade library.

CONTENT DETAILS

MODULE I SECONDARY AND AUXILIARY MOTIONS.

Secondary motions - take up motions - 7- wheel take up motion - continuous and intermittent take up motion- Calculations related to take up motions-Let off motions- working of a negative let off motion-working of positive let off motions(Ruti – Ropper)-auxiliary motions–Side weft fork and centre weft fork motions-working of side weft fork motion with sketch--timing and setting of side weft fork motion-objectives and types of warp stop motions - working of a mechanical warp stop motion used in 'Sakamoto' looms with neat sketch-working of electrical warp stop motion with neat sketch-objectives and types of warp protector motions- working of loose reed motion with neat sketch-the working of fast reed motion with neat sketch-the objectives of brake motions-the working of brake motion with neat sketch-the objectives and general description of check strap- the objectives and general description of Temples

MODULE II DOBBY SHEDDING.

Various classification of dobby shedding mechanisms- dobby designs- method of pegging of lags for L.H and R.H dobbies – working of Keighley dobby and its limitations-timing and setting of keighley dobby- advantages of double jack dobby over single jack dobby-working of Climax dobby- -use of Cross Border dobby-working of Cross border dobby-working of Ruti Cam dobby -working of Knowles positive open shed dobby-functions of heald leveling device and steadying mechanism in dobby.

MODULE III JACQUARD SHEDDING.

Advantages of Jacquard shedding-basic principle of Jacquard shedding-list the parts & functions of jacquard -harness mounting like London and Norwich systems- define harness ties and various types of harness ties like straight tie, centre tie, mixed tie- count of comber board - First hook and First needle of the jacquard - various classifications of Jacquards - Single lift Jacquards, Double Lift Jacquards, Single Cylinder Jacquards, Double Cylinder Jacquards, Cross border Jacquard - working of Single Lift Jacquards-working of Double Lift Single Cylinder Jacquard-working of Double Lift Double Cylinder Jacquard-working of Cross Border Jacquard- -construction and development of Jacquard designs-working of Piano card cutting machine-card punching and card lacing-use of special Jacquards and the working of Pressure Harness.

MODULE IV COMPUTER AIDED TEXTILE DESIGNING.

Introduction of textile CAD for woven designs – common features of textile CAD - creation of weave, draft and peg plan by using CAD - Creation of yarn library for different counts, type and quality- Creation of library of different weaves - creation of draft and peg plan from weaves - produce stripe, cross over and check designs.- printed design and its colour Separations. - fabric simulation-

steps to produce a dobby designs - steps & commands to produce a Jacquard design - produce fabric simulation in different Colour combinations-take graph print outs as per the requirements of loom - a texture mapping on various objectives and models with new designs - colour library- shade library.

REFERENCE BOOKS

1.Thomas W. Fox	The Mechanism of Weaving	Universal Publishing corp. Mumbai -2
2 Marks & Robinson(ATC)	Principles of Weaving	The Textile Institute, Manchester.
3 N.N.Banerjee	Weaving Mechanism Vol – I	Smt Tandra banerjee & Smt Apurba banerjee ,West Bengal
4 NCUTE	Woven Fabric Production II	NCUTE New Delhi
5 R.Sengupta	Weaving Calculations	D.B.Taraporevala sons & coLtd., Mumbai
6 A.Ormerod	Modern Preparation and Weaving	Wood Head Publishing Ltd,London
7 Prof. J.C.Chakravorthy	Mechanism of Weaving Vol- I&II	Prof. J.C.ChakravorthyWest Bengal
8 B.Hasmukrai	Fabric Forming	SSMITT&PC Co-OperativeStores